

G Rainbow

A drone organ with a two-deck time-lag tape-loop delay, after the 60s modified-organ and tape-loop pieces.



Overview

G Rainbow is an eight-voice organ built for drones and minimalist loops. Each voice has two oscillator ranks, each tuned to a modal interval of the note you play, with a wave that blends smoothly from sine through triangle and saw to pulse. The ranks go through a saturating four-pole ladder filter and a simple organ-style amp envelope with vibrato.

The heart of the instrument is the TIME-LAG ACCUMULATOR: two virtual tape decks in series. The organ is recorded on deck 1, deck 1's playback feeds deck 2, and deck 2's playback is fed back into deck 1's record head - so what you play piles up, layer on layer, slowly darkening and wobbling like old tape. HOLD freezes the loop so you can play over it.

A drift control adds a slow, audible wander of pitch, detune, pulse width, filter and tape speed, plus a small random tuning on every note.

Reach for it for

- Evolving drones and pads
- Minimalist repeating patterns that build up over time
- Tape-loop textures to play over (HOLD)

- Detuned organ tones at modal intervals
- Ambient beds that keep moving

Getting started

1. Insert G Rainbow on a track (it is an instrument: it has no audio input).
2. Arm the track with a MIDI input, or draw notes in a MIDI item. It plays on any MIDI channel and responds to velocity and the sustain pedal (CC64).
3. Pick a preset from the display at the top, then play a few notes and let the loops build.

Controls

Knobs: drag up or down (Shift for fine), double-click or Ctrl-click to reset. A knob's value shows while you hover over it, and the display at the top briefly shows the control you touched.

The header

Preset display	Shows PRESET nn/16 and the name. Click it for the list of presets; the arrows either side step through them.
SIZE (S / M / L)	Panel size: small, medium or large.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
PANIC	Releases every note.
MIDI light	Flashes when MIDI arrives.
HW	Switches between the detailed hardware look and a flat look.
COBALT	Switches the panel from copper to a complementary blue colour scheme.

RANK A and RANK B

Interval	The arrows choose where the rank sounds relative to the note you play: -2 OCT, -1 OCT, UNISON, MIN 3RD, MAJ 3RD, 4TH, 5TH, OCTAVE, 12TH or 2 OCT.
Wave scope	Two cycles of the rank's current waveform.
WAVE	Blends the waveform from sine through triangle and saw to pulse, 0 to 100%. The label shows the dominant wave.
LEVEL	The rank's level, 0 to 100%.
DETUNE	Fine tuning of the rank, -50 to +50 cents. A few cents between the ranks gives a slow beating.

LADDER FILTER

CUTOFF	40 Hz to 16 kHz. It follows the keyboard by half, so higher notes are a little brighter.
RESO	Resonance, 0 to 100%. Near the top the filter starts to self-oscillate.
DRIVE	How hard the ranks drive the filter, 0 to 100%, for a thicker, saturated tone.
Response display	A sketch of the filter's frequency response.

AMP

ATTACK	1 ms to 3 s.
RELEASE	10 ms to 8 s.
VIBRATO	Organ-style vibrato at about 5.8 Hz, 0 to 100% (up to 30 cents).
Envelope display	A sketch of the attack and release.

TIME-LAG ACCUMULATOR

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DECK 1 / DECK 2 (reels)	The two decks. The readout shows each deck's length, and the bar under the reels shows the loop position.
DECK 1	The length of deck 1's loop, 0.5 to 20 seconds. Click the DECK 1 label under the knob to switch to bars (1/2, 1, 2, 3, 4, 6, 8 or 12 bars of the project tempo, up to 20 seconds).
DECK 2	The length of deck 2's loop, set the same way. Deck 2 plays deck 1's output a further DECK 2 later.
FEEDBACK	How much of deck 2's playback is re-recorded onto deck 1 on every pass, 0 to 100%. High settings let the layers pile up for a long time.
AGE	Tape wear on every pass, 0 to 100%: the repeats get darker and more saturated (the lows below about 35 Hz are kept clean).
WOW	Wow and flutter of the transports, 0 to 100%: a slow pitch wander with a little fast flutter.
MIX	The level of the loops against the organ, 0 to 100%. The left side carries more of deck 1, the right more of deck 2.
HOLD	Freezes the loop: new playing is no longer recorded and what is on the tape keeps circulating (almost without loss). Play over it; switch HOLD off to record again.

MASTER

DRIFT	0 to 100% (30% by default). A slow wander of pitch (up to 8 cents), extra detune on rank B, pulse width, the filter cutoff and the tape speed, plus a random tuning offset on every new note (up to 5 cents).
OUTPUT	Output level, -24 to +6 dB. A built-in limiter keeps the output just under full scale.
OUT meter	Output level.

Presets

Every preset sets the ranks, filter, amp and accumulator, and resets DRIFT to 30% and OUTPUT to -6 dB. HOLD is switched off.

Clean Organ	Unison and octave ranks, the accumulator off: the plain organ.
Rainbow	Unison and octave with 4 s and 6 s decks building up at 60% feedback.
Dervish	Unison and fifth, darker and more resonant, with deeper vibrato, more feedback, age and wow.
Poppy	A bright pulse-leaning unison rank with a 12th above, strong vibrato and long 8 s / 12 s decks.
In C Pulse	Short, plucky organ notes with the decks synced to 1 and 2 bars, for repeating patterns.
Keyboard Study	A brighter, softer organ with short 1.5 s / 2.5 s loops.
Copper Cathedral	An octave below and two octaves above, slow attack and release, long 12 s / 18 s decks at 80% feedback.
Rust Drone	An octave below and a fifth, dark and heavily driven, with lots of age and wow.
Sunrise Drone	Soft waves an octave below and a fifth, a slow 800 ms attack and no vibrato.
Reed Swarm	Buzzy unison and octave ranks, strong vibrato and short 2 s / 3 s loops.
Tape Echoes	Unison and a 12th with short 0.75 s / 1.25 s decks, for echoes rather than layers.
Modal Fifths	Unison and fifth with 5 s / 7.5 s decks.
Glass Harmonics	Near-sine ranks at an octave and two octaves, an open filter and a slow attack.

Minor Third Mist	Unison and a minor third, a slow attack and well-aged 7 s / 11 s loops.
Pulse Machine	Very short, resonant notes with the decks synced to half a bar and one bar.
Frozen Choir	Unison and fifth, a 1.2 s attack and 6 s release, long 15 s / 20 s decks at 90% feedback.

Tips

- Make the two deck lengths an uneven ratio (4 s and 6 s, 5 s and 7.5 s) so the layers keep shifting against each other rather than lining up.
- Build a bed with FEEDBACK high, then press HOLD and play a melody over the frozen loop.
- Sync the decks to bars when you want the loops to stay in time with the project.
- If the loops get too dark or muddy, lower AGE; if they get too loud, lower FEEDBACK or MIX.

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